

Chapter 16: String Theory and Matrix Models

1. The Nambu-Goto action is

$$-T_2 \int d^3\xi [\sqrt{-\det \hat{g}} + \hat{C}_{\alpha\beta\gamma} \epsilon^{\alpha\beta\gamma}], \quad \hat{g}_{\alpha\beta} = G_{\mu\nu} \partial_\alpha X^\mu \partial_\beta X^\nu, \quad C_{\alpha\beta\gamma} = C_{\mu\nu\rho} \partial_\alpha X^\mu \partial_\beta X^\nu \partial_\gamma X^\rho$$

Let's set $C_{\alpha\beta\gamma} = 0$. The EOM for the scalar field is quickly seen to be $\square X = 0$, where $\square$ is the Laplacian from the induced metric.

In the Polyakov action, the equations of motion for γ are the vanishing the energy-momentum tensor, giving:

$$\partial_\alpha X^\mu \partial_\beta X_\mu - \frac{1}{2} \gamma_{\alpha\beta} (\gamma^{\gamma\delta} \partial_\gamma X^\mu \partial_\delta X_\mu - 1)$$

This is harder to solve than the $p = 1$ case, as we can't just take the square root of the determinant of both sides. Taking the ansatz that $\gamma_{\alpha\beta} = \lambda \partial_\alpha X^\mu \partial_\beta X_\mu$ we get:

$$\frac{1}{\lambda} \gamma_{\alpha\beta} - \frac{1}{2} \gamma_{\alpha\beta} \left(\frac{3}{\lambda} - 1 \right)$$

And we get a solution with $\lambda = 1$. Note there is no Weyl rescaling here. Similarly, the X field must satisfy

$$\frac{1}{\sqrt{-h}} \partial_\alpha (\sqrt{-h} h^{\alpha\beta} \partial_\beta X^\mu) = 0$$

which agrees with $\square X = 0$ upon the identification of the induced and auxiliary metrics.

Note importantly that the p -brane action for $p \neq 1$ requires a cosmological constant term.

2. Take the gauge $\gamma_{00} = -\det \hat{g}_{ij}$ with $\gamma_{0i} = 0$ so that $\sqrt{-\gamma} = \det g_{ij}$. The action then becomes:

$$-\frac{T_2}{2} \sqrt{\gamma} (\gamma^{00} \dot{X} \cdot \dot{X} - 1) = \frac{T_2}{2} (\dot{X} \cdot \dot{X} + \det \hat{g}_{ij})$$

Here there is a small typo in Kiritsis. Rewriting

$$\det \hat{g}_{ij} = \partial_1 X^\mu \partial_1 X^\nu \partial_2 X_\mu \partial_2 X_\nu - \partial_1 X^\mu \partial_2 X^\nu \partial_1 X_\mu \partial_2 X_\nu = -\frac{1}{2} \{X^\mu, X^\nu\} \{X_\mu, X_\nu\}$$

We thus get total action:

$$\frac{T_2}{2} \int d^3\xi \left(\dot{X}^\mu \dot{X}_\mu - \frac{1}{2} \{X^\mu, X^\nu\} \{X_\mu, X_\nu\} \right)$$

Giving equations of motion

$$\ddot{X}^\mu = \{ \{X^\mu, X^\nu\}, X_\nu \}$$

Taking now lightcone gauge $X^+(\tau, \sigma_1, \sigma_2) = \tau$

The transverse momenta are:

$$p^i = \frac{\delta L}{\delta(\partial_\tau X^i)} = T_2 \dot{X}^i = \frac{p^+}{V} \dot{X}^i$$

The Hamiltonian is thus

$$p_- \dot{X}^- - \mathcal{L} + \int d^2\xi p_i \dot{X}^i$$

3. This rescaling is very straightforward once one has the hamiltonian 16.1.14. One rescales $X^i \rightarrow \left(\frac{N}{V_2}\right)^{1/4} X^i$ and $t \rightarrow \left(\frac{N}{4V_2}\right)^{-1/4} t$ yielding:

$$\frac{T_2}{2} \int d^2\sigma \dot{X}^i \dot{X}^i \rightarrow \frac{T_2}{4} \frac{N}{V_2} \int d^2\sigma \frac{\dot{X}^i \dot{X}^i}{2} = \frac{T_2}{4} \text{Tr} \left[\frac{\dot{X}^i \dot{X}^i}{2} \right]$$

and

$$\frac{T_2}{4} \int d^2\sigma \{X^i, X^j\} \{X_i, X_j\} \rightarrow -\frac{T_2}{4} \frac{N}{V_2} \int d^2\sigma \frac{1}{4} [X^i, X^j] [X_i, X_j] = \frac{T_2}{4} \text{Tr} \left(-\frac{1}{4} [X^i, X^j] [X_i, X_j] \right)$$

4. For a string, imagine a rectangular spike of cross-section ϵ and length L . Its total energy is $2L + \epsilon$, where ϵ does not multiply L now. Therefore, taking L large will give a large energy deviation, regardless of how small we take ϵ . Thus, the string is stable against decaying into these small spikes.
5. Its immediate that the $C_{\mu\nu\rho}$ term multiplies a Nambu bracket, by antisymmetry. Now by permutation invariance we can write:

$$\frac{1}{6}\{X^\mu, X^\nu, X^\rho\}\{X_\mu, X_\nu, X_\rho\} = \partial_1 X^\mu \partial_2 X^\nu \partial_3 X^\rho \epsilon_{\alpha\beta\gamma} (\partial_\alpha X_\mu \partial_\beta X_\nu \partial_\gamma X_\rho)$$

Its not hard to see that this reproduces the formula for a 3x3 determinant, as we have an antisymmetric object involving one element from every row and column multiplied together, all with unital coefficients. That is, $\frac{1}{6}\{X^\mu, X^\nu, X^\rho\}\{X_\mu, X_\nu, X_\rho\} = \det \hat{g}_{\alpha\beta}$, which is negative for Lorentzian signature, hence the minus sign under the square root in the bracket form of the action: $\sqrt{-\det \hat{g}} = \sqrt{-\frac{1}{6}\{X^\mu, X^\nu, X^\rho\}\{X_\mu, X_\nu, X_\rho\}}$.

The bracket is not associative **show**